“mathematics is a wonderful gift we neither understand nor deserve.”

---------- Eugene Wigner, who won the Nobel Prize for introducing group theory into quantum mechanics…



- **Trust me, it's not that hard**

In 1951, I had the good fortune of listening to Professor Racah's lecture on Lie groups at Princeton. After attending these lectures, I thought, "This is really too hard. I cannot learn all this . . . too damned hard and unphysical."

—A. Salam, 1979 Nobel laureate in physics

Trust me, it's not that hard. And as Salam's own Nobel-winning work helped show, group theory is certainly relevant to physics. We now know that the interweaving gauge bosons underlying our world **dance** to the music of Lie groups and Lie algebras.

------ A. Zee , preface in 《group theory》

“Numbers have a way of taking a man by the hand and leading him down the path of reason."

-------- PyThagoras

Nous sommes des enfants, mais les enfants bien courageux et pleins en forme.

我们是孩子，但我们精力充沛，勇往直前。。。

---------- Évariste Galois

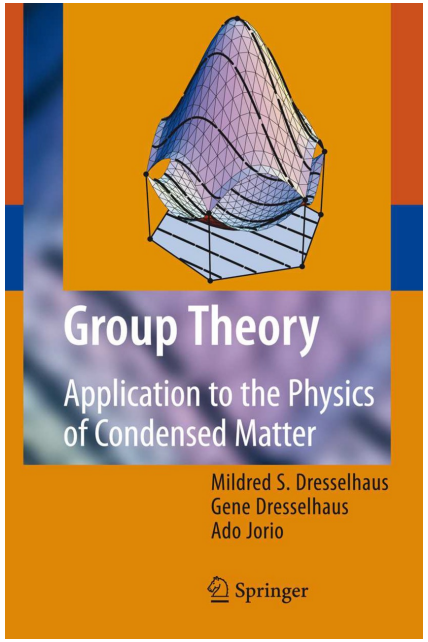
Galois是Grothendieck（《Récoltes et Semailles》：

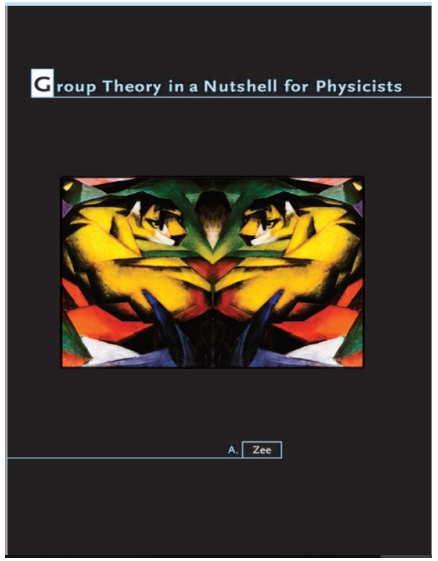
收获与播种）最喜爱的数学家

References:

Group theory
Application to the physics of condensed matter
Mildred S. Dresselhaus
Gene Dresselhaus
Ado Jorio
Springer press

Group theory in a nutshell for physicists
A.Zee, Princeton university press





Quantum mechanics is linear

$\mathcal{H}_0$: unperturbed Hamiltonian matrix

$\mathcal{H}'$: some interaction Hamiltonian matrix

ψ_α , ψ_β : eigenfunctions for $\mathcal{H}_0$

$\left(\psi_\alpha,\ \mathcal{H}'\psi_\beta\right)$ is a number, will it transform as the fully symmetric representation Γ_1 ?

If not , it vanishes! It is not the scalar in the representation.

在考虑物理问题中的各种选择定则时，我们经常需要考虑 降低未扰问题的对称性的微扰哈密顿量的矩阵元。

存在电磁场的哈密顿量可以写为 $\mathcal{H}=\frac{1}{2m}(\vec{p}-\frac{e}{c}\vec{A})^2+V$

则固体中存在电磁场时电子的哈密顿量为

$$\mathcal{H}=\frac{(\vec{p}-\frac{e}{c}\vec{A})^2}{2m}+V(\vec{r})$$

已选库伦规范

$$=\frac{p^2}{2m}+V(\vec{r})-\frac{e}{mc}\vec{A}.\vec{p}+\frac{e^2A^2}{2mc^2}$$

其中$\vec{A}$是电磁场的矢势，$V(\vec{r})$是周期势。因此无电磁场的单电子哈密顿量为

$$\mathcal{H}_0=\frac{p^2}{2m}+V(\vec{r})$$

电磁微扰项

$$\mathcal{H}'_{em}=-\frac{e}{mc}\vec{A}.\vec{p}+\frac{e^2A^2}{2mc^2}$$

一般起主要作用的是第一项

$$\mathcal{H}'_{em}\cong-\frac{e}{mc}\vec{A}.\vec{p}$$

本征函数ψ_β选未扰哈密顿量$\mathcal{H}_0$的某不可约表示Γ_i的基函数分量$\psi_j^{(\Gamma_i)}$

outline

这章讨论的选择定则围绕以下课题组织

基函数的对称性规则(a)
群和表示的直积(b)
用群论术语表达选择定则概念(c)
举例(d)

基函数的正交性

$\psi_\alpha^{(i)}$ —— 不可约表示Γ_i

$\psi_\alpha^{(i)}$ —— 分量

$$\hat{P}_R\psi_\alpha^{(i)} = \sum_{j=1}^{l_i} D^{(i)}(R)_{j\alpha}\psi_j^{(i)}$$

其中$\hat{P}_R$是对称算符，$\psi_\alpha^{(i)}$标记l_i维不可约表示(i)的基函数，$D^{(i)}(R)_{j\alpha}$是不可约表示(i)中对称群元R的矩阵元。

定理：属于不同的不可约表示或属于同一不可约表示的两个基函数是正交的。

证明:

令$\phi_{\alpha}^{(i)}$和$\psi_{\alpha'}^{(i')}$分别属于不可约表示(i)和(i')并相应于它们各自表示的第α列和α'列。

根据定义

$$\hat{P}_R \phi_{\alpha}^{(i)} = \sum_{j=1}^{l_i} D^{(i)}(R)_{j\alpha} \phi_j^{(i)}$$

$$\hat{P}_R \psi_{\alpha'}^{(i')} = \sum_{j'=1}^{l_{i'}} D^{(i')}(R)_{j\alpha'} \psi_{j'}^{(i')}$$

因为标量积（或两个态之间的单位算符的矩阵元）独立于坐标系，我们可以将标量积写为

$$(\phi_{\alpha}^{(i)}, \psi_{\alpha'}^{(i)}) = (\hat{P}_R \phi_{\alpha}^{(i)}, \hat{P}_R \psi_{\alpha'}^{(i)}) = \sum_{jj'} D^{(i)}(R)_{j\alpha}^{*} D^{(i)}(R)_{j\alpha'} (\phi_j^{(i)}, \psi_{j'}^{(i)})$$

由于$(\phi_{\alpha}^{(i)}, \psi_{\alpha'}^{(i)})$显然不依赖于群元$R$，所以

$$(\phi_{\alpha}^{(i)} \psi_{\alpha'}^{(i')}) = \frac{1}{h} \sum_{R} (\phi_{\alpha}^{(i)} \psi_{\alpha'}^{(i)})$$

$$= \frac{1}{h} \sum_{jj'} \sum_{R} D^{(i)}(R)_{j\alpha}^{*} D^{(i)}(R)_{j\alpha'} (\phi_{j,}^{(i)}, \psi_j^{(i)})$$

其中h为群阶。

现在应用绝妙正交性定理(Van Vleck 命名),

$$\frac{1}{h}\sum_{R} D^{(i)}(R)^{*}_{j\alpha} D^{(i')}(R)_{j'\alpha'} = \frac{1}{l_i}\delta_{ii'}\delta_{jj'}\delta_{\alpha\alpha'}$$

代入上式得

$$(\phi_{\alpha}^{(i)}, \psi_{\alpha'}^{(i')}) = \frac{1}{l_i}\delta_{ii'}\delta_{\alpha\alpha'}\sum_{jj'}\delta_{jj'}(\phi_{j}^{(i)}, \psi_{j'}^{(i')})$$

$$= \frac{1}{l_i}\delta_{ii'}\delta_{\alpha\alpha'}\sum_{j=1}^{l_i}(\phi_{j}^{(i)}, \psi_{j}^{(i)})$$

因此，如果基函数$\phi_{\alpha}^{(i)}$和$\psi_{\alpha'}^{(i')}$相应于两个不同的不可约表示$i \neq i'$，它们是正交的；如果它们相应于相同的表示$i = i'$，但如果它们相应于矩阵的不同列（行），也就是它们相应于表示i的不同分量$\alpha \neq \alpha'$，则也正交。

我们进而注意到上式右边不依赖于α（当$\alpha = \alpha'$时），因此这个标量积对所有的分量α都相同。

$$\mathcal{H}_0\psi_{\alpha'}^{(i')} = E_{\alpha'}^{(i')}\psi_{\alpha'}^{(i')}$$

如果$\phi_{\alpha}^{(i)}$和$\psi_{\alpha'}^{(i')}$都是哈密顿量$\mathcal{H}_0$的本征函数，且也是不可约表示(i)和(i')的基函数，则矩阵元$(\phi_{\alpha}^{(i)}, \mathcal{H}_0\psi_{\alpha'}^{(i')})$在$i \neq i'$或$\alpha \neq \alpha'$时为零。

两个不可约表示的直积

对于直积群，我们可以用两个矩阵的直积很方便定义两个表示的直积。

由代数知，我们定义两个矩阵的直积$A \otimes B = C$为A的每个矩阵元乘以 B的每个矩阵元，因此直积矩阵C就有两套指标：

$$A_{ij}B_{kl} = C_{ik,jl}$$

因此，如果A是一个2×2的矩阵， B是一个3×3的矩阵，则C是6×6的矩阵。

定理：群A和群B的表示的直积构成了其直积群的表示。

证明：
我们需要证明

$$D_{ij}^{(a)}(A_i)D_{pq}^{(b)}(B_j) = (D^{(a\otimes b)}(A_iB_j))_{ip,jq}$$

为了证明这个定理 我们需要证明

$$D^{(a\otimes b)}(A_kB_l)D^{(a\otimes b)}(A_{k'}B_{l'}) = D^{(a\otimes b)}(A_iB_j)$$

其中$A_i = A_kA_{k'}, B_j = B_lB_{l'}$，
根据直积群的定义，群A的群元和群B的群元对易，则直积群的乘法规则为

$$A_kB_lA_{k'}B_l = A_kA_{k'}B_lB_{l'} = A_iB_j$$

其中是A_kB_l是直积群中的典型群元，我们现在必须证明表示也遵守这一乘法规则。

根据定义

$$D^{(a\otimes b)}(A_kB_l)D^{(a\otimes b)}(A_{k'}B_{l'})$$
$$= \quad [D^{(a)}(A_k)\otimes D^{(b)}(B_l)][D^{(a)}(A_{k'})\otimes D^{(b)}(B_{l'})]$$

$$= \sum_{sr} D^{(a)}(A_k)_{is}D^{(b)}(B_l)_{pr}D^{(a)}(A_{k'})_{sj}D^{(b)}(B_{l'})_{rq}$$
$$= \sum_{s} D^{(a)}(A_k)_{is}D^{(a)}(A_{k'})_{sj}\sum_{r} D^{(b)}(B_l)_{pr}D^{(b)}(B_{l'})_{rq}$$
$$= D^{(a)}(A_kA_{k'})_{ij}D^{(b)}(B_lB_{l'})_{pq}$$
$$= D^{(a)}(A_i)_{ij}D^{(b)}(B_j)_{pq}$$
$$= D^{(a\otimes b)}(A_iB_j)_{ip,jq} = \sum_{sr}(D^{(a)}(A_k)\otimes D^{(b)}(B_l))_{ip,sr}(D^{(a)}(A_{k'})\otimes D^{(b)}(B_{l'})_{sr,jq}$$
$$\Rightarrow [D^{(a\otimes b)}(A_kB_l)D^{(a\otimes b)}(A_{k'}B_{l'})]_{ip,jq}$$

$$= D^{(a)}(A_kA_{k'})_{ij}D^{(b)}(B_lB_{l'})_{pq}$$
$$= D^{(a)}(A_i)_{ij}D^{(b)}(B_j)_{pq}$$
$$= D^{(a\otimes b)}(A_iB_j)_{ip,jq}$$

直积的特征标

定理：直积群的特征标或两个表示取直积的特征标的可以想象的最简单公式为：
(a)如果对两个群取直积，则直积群的不可约表示的特征标可由原群的不可约表示的特征标乘积得到。

$$\chi^{(a\otimes b)}(A_kB_l)=\chi^{(a)}(A_k)\chi^{(b)}(B_l)$$

(b)如果对同群的两个表示取直积，则直积表示的特征标写为

$$\chi^{(l_1\otimes l_2)}(R)=\chi^{(l_1)}(R)\chi^{(l_2)}(R)$$

证明：
考虑直积群中一个群元的对角矩阵元。由两个群的直积的定义有：

$$D^{(a\otimes b)}(A_kB_l)_{ip,jq}=D^{(a)}(A_k)_{ij}D^{(b)}(B_l)_{pq}$$

取上式的对角元并求和得

$$\sum_{ip}D^{(a\otimes b)}(A_kB_l)_{ip,ip}=\sum_{i}D^{(a)}(A_k)_{ii}\sum_{p}D^{(b)}(B_l)_{pp}$$

写成迹的表达式：

$$\chi^{(a\otimes b)}(A_kB_l)=\chi^{(a)}(A_k)\chi^{(b)}(B_l)$$

这个结果对类也成立。
对同一个群的两个表示的直积的特征标的证明也是类似的。

一般来说，如果我们取一个群的两个不可约表示的直积，则相应的直积表示是可约的。并且如果是可约的，则直积的特征标可以写为群的不可约表示的特征标的线性组合：

$$\chi^{(\lambda)}(R)\chi^{(\mu)}(R) = \sum_{\nu} a_{\lambda\mu\nu}\chi^{(\nu)}(R)$$

从特征标的分解公式:

$$a_j = \frac{1}{h}\sum_{k} N_k[\chi^{(\Gamma_j)}(C_k)]^*\chi(C_k)$$

得

$$a_{\lambda\mu\nu} = \frac{1}{h}\sum_{C_\alpha} N_{C_\alpha}\chi^{(\nu)}(C_\alpha)^*[\chi^{(\lambda)}(C_\alpha)\chi^{(\mu)}(C_\alpha)]$$

其中， C_α表示类， N_{C_α}表示类C_α中群元的个数。

首先，我们找出直积群的群元和类，然后利用 G_A和G_B的特征标表构造$G_A \otimes G_B$的特征标表。我们以$C_{4h} = C_4 \otimes i$为例。

Table 6.1. Character table for point group C_{4h}

	$C_{4h} \equiv C_4 \otimes i$				$(4/m)$				
	E	C_2	C_4	C_4^3	i	iC_2	iC_4	iC_4^3	
A_g	1	1	1	1	1	1	1	1	
B_g	1	1	−1	−1	1	1	−1	−1	even under
E_g	1	−1	i	$-i$	1	−1	i	$-i$	inversion (g)
	1	−1	$-i$	i	1	−1	$-i$	i	
A_u	1	1	1	1	−1	−1	−1	−1	
B_u	1	1	−1	−1	−1	−1	1	1	odd under
E_u	1	−1	i	$-i$	−1	1	$-i$	i	inversion (u)
	1	−1	$-i$	i	−1	1	i	$-i$	

特征标表

C_i	E	i
	1	1
	1	−1

C_4是不变子群：***i***
作用后其表示的符号不变，偶宇称；
符号改变，奇宇称。

C_{4h}的特征标表中，我们利用g表示空间反演偶宇称，利用 u表示空间反演奇宇称。我们注意到左上方的四分之一表是C_4群的特征标，右边的四个类是把C_4的四个类乘以i得到的。右上方和左下方的 四分之一表是一样的，而右下方的表是把C_4的特征标全部乘以(-1)来获得C_{4h}群的奇不可约表示。

令波函数$\phi_{\alpha}^{(i)}$和$\psi_{\alpha'}^{(i')}$分别随Γ_i和$\Gamma_{i'}$不可约表示的分量α和α'变换，并令$\mathcal{H}'$随表示$\boldsymbol{\Gamma_j}$**变换(?)**。如果直积$\Gamma_j \otimes \Gamma_i$与$\Gamma_{i'}$正交，则矩阵元为零；或者等价地说，如果$\Gamma_{i'} \otimes \Gamma_j \otimes \Gamma_i$不包含完全对称表示$\Gamma_1$（恒等表示），矩阵元为零。

更一般的情形，我们用薛定谔方程群中所包含的$\mathcal{H}'$的不可约表示 表达$\mathcal{H}'$：

$$\mathcal{H}' = \sum_{j\beta} f_{\beta}^{(j)} \mathcal{H}_{\beta}'^{(j)}$$

其中j标记哈密顿量$\mathcal{H}'$的不可约表示Γ_j，β标记Γ_j 的分量。 $\mathcal{H}'\phi_{\alpha}^{(i)}$随$\mathcal{H}_{\beta}'^{(j)} \otimes \phi_{\alpha}^{(i)}$直积形成的直积表示而变换，其中$i$标记不可约表示$\Gamma_i$， 用对称性的标记则为$\Gamma_{j,\beta} \otimes \Gamma_{i,\alpha}$。

矩阵元$\left(\boldsymbol{\psi}_{\alpha'}^{(i')}, \mathcal{H}'\boldsymbol{\phi}_{\alpha}^{(i)}\right)$消失，当且仅当$\boldsymbol{\psi}_{\alpha'}^{(i')}$与$\mathcal{H}'\boldsymbol{\phi}_{\alpha}^{(i)}$分解为不可约表示的所有基函数正交。

同概念的一个等价的表达为$\psi_{\alpha'}^{(i')} \otimes \mathcal{H}_{\beta}'^{(j)} \otimes \phi_{\alpha}^{(i)}$为了令$\left(\psi_{\alpha'}^{(i')}, \mathcal{H}'\phi_{\alpha}^{(i)}\right)$中的矩阵元为非零，这个三重直积必须包含随一个标量或一个常数变换的项，或者是根据不可约表示Γ_1变换的项。

$\boldsymbol{\mathcal{H}'}$随表示$\boldsymbol{\Gamma_j}$变换?

假设$\boldsymbol{\Gamma_j}$表示为二维表示，这个二维空间的维度源自于$\mathcal{H}_0$的一个对称群（$\{R_1 = E, R_2\}$ 两个群元），
其能级二重简并。 $\boldsymbol{\mathcal{H}'}$是二阶张量，

$$\widetilde{\boldsymbol{\mathcal{H}}'}_{ij} = \hat{P}_R\, \boldsymbol{\mathcal{H}'}_{ij} = D_{ik}^{(\boldsymbol{\Gamma_j})}(R) D_{jl}^{(\boldsymbol{\Gamma_j})}(R) \boldsymbol{\mathcal{H}'}_{kl}$$

$$\hat{P}_R \phi_\alpha^{(\boldsymbol{\Gamma_j})} = \sum_{j=1}^{2} D^{(\boldsymbol{\Gamma_j})}(R)_{j\alpha} \phi_j^{(i)}$$

取一个矩阵元$\widetilde{\boldsymbol{\mathcal{H}}'}_{12}$

$$\widetilde{\boldsymbol{\mathcal{H}}'}_{12} = D_{11}^{(\boldsymbol{\Gamma_j})}(R) D_{21}^{(\boldsymbol{\Gamma_j})}(R) \boldsymbol{\mathcal{H}'}_{11} + D_{11}^{(\boldsymbol{\Gamma_j})}(R) D_{22}^{(\boldsymbol{\Gamma_j})}(R) \boldsymbol{\mathcal{H}'}_{12} + D_{12}^{(\boldsymbol{\Gamma_j})}(R) D_{21}^{(\boldsymbol{\Gamma_j})}(R) \boldsymbol{\mathcal{H}'}_{21} + D_{12}^{(\boldsymbol{\Gamma_j})}(R) D_{22}^{(\boldsymbol{\Gamma_j})}(R) \boldsymbol{\mathcal{H}'}_{22}$$

…

$$\widetilde{\boldsymbol{\mathcal{H}}'} = D^{(\boldsymbol{\Gamma_j})} \otimes D^{(\boldsymbol{\Gamma_j})} \boldsymbol{\mathcal{H}'} = D^{(\boldsymbol{\Gamma_j})} \otimes D^{(\boldsymbol{\Gamma_j})} \begin{pmatrix} \boldsymbol{\mathcal{H}'}_{11} \\ \boldsymbol{\mathcal{H}'}_{12} \\ \boldsymbol{\mathcal{H}'}_{21} \\ \boldsymbol{\mathcal{H}'}_{22} \end{pmatrix}$$；相同表示的直积： 4×4 表示矩阵

$$4 = 2 \times 2$$

选择定则的例子

我们现在考虑在一个有O_h对称性的系统中的电偶极跃迁，导致电偶极跃迁的电磁相互作用为

$$\mathcal{H}'_{em} = -\frac{e}{mc}\vec{A}.\vec{p}$$

其中$\vec{A}$是外加电磁场的矢势，$\vec{p}$是电子的动量。这个动量算符是所考虑的电子系统的部分，而电磁场的矢量$\vec{A}$则起到外加系统的作用。或者在热力学的意义下作为bath或源(reservoir)。

$\vec{p}$是与薛定谔方程群相关的一个算符，但是$\vec{A}$在薛定谔方程群的对称操作下(作为欧氏空间中的三个分量：A_x, A_y, A_z 三个数)保持不变（与薛定谔方程的希尔伯特空间变换无关），不随其变换。因此，用群论的语言，在未扰系统$\mathcal{H}_0\psi = E\psi$的薛定谔方程群的语境下，电磁相互作用随着一个矢量（$\vec{p}$）而变换。在立方对称性中，矢量的所有三个分量都随相同的不可约表示而变换；如果我们把系统换成四方对称性，则p_x和p_y会随某二维不可约表示变换，而 p_z会随某一维不可约表示变换。

Table A.30. Character table for group O (cubic)

O (432)		E	$8C_3$	$3C_2 = 3C_4^2$	$6C_2'$	$6C_4$
$(x^2+y^2+z^2)$	A_1	1	1	1	1	1
	A_2	1	1	1	−1	−1
$(x^2-y^2, 3z^2-r^2)$	E	2	−1	2	0	0
(R_x, R_y, R_z) / (x, y, z)	T_1	3	0	−1	−1	1
(xy, yz, zx)	T_2	3	0	−1	1	−1

$O_h = O \otimes i$, $(m3m)$ (cubic)

Table A.31. Character table for the cubic group O_h (cubic)†

repr.	basis functions	E	$3C_4^2$	$6C_4$	$6C_2'$	$8C_3$	i	$3iC_4^2$	$6iC_4$	$6iC_2'$	$8iC_3$
A_1^+	1	1	1	1	1	1	1	1	1	1	1
A_2^+	$x^4(y^2-z^2)+$ $y^4(z^2-x^2)+$ $z^4(x^2-y^2)$	1	1	−1	−1	1	1	1	−1	−1	1
E^+	x^2-y^2 / $2z^2-x^2-y^2$	2	2	0	0	−1	2	2	0	0	−1
T_1^-	x, y, z	3	−1	1	−1	0	−3	1	−1	1	0
T_2^-	$z(x^2-y^2)\ldots$	3	−1	−1	1	0	−3	1	1	−1	0
A_1^-	$xyz[x^4(y^2-z^2)+$ $y^4(z^2-x^2)+$ $z^4(x^2-y^2)]$	1	1	1	1	1	−1	−1	−1	−1	−1
A_2^-	xyz	1	1	−1	−1	1	−1	−1	1	1	−1
E^-	$xyz(x^2-y^2)\ldots$	2	2	0	0	−1	−2	−2	0	0	1
T_1^+	$xy(x^2-y^2)\ldots$	3	−1	1	−1	0	3	−1	1	−1	0
T_2^+	xy, yz, zx	3	−1	−1	1	0	3	−1	−1	1	0

† The basis functions for T_2^- are $z(x^2-y^2)$, $x(y^2-z^2)$, $y(z^2-x^2)$, for E^- are $xyz(x^2-y^2)$, $xyz(3z^2-s^2)$ and for T_1^+ are $xy(x^2-y^2)$, $yz(y^2-z^2)$, $zx(z^2-x^2)$

在立方体群O_h中，**矢量(x, y, z)随不可约表示T_{1u}变换**。(p_x, p_y, p_z)也一样。因为它们都是极矢量，在空间反演下为奇宇称。

O_h群有10个不可约表示，5个偶宇称，5个奇宇称。

对于偶宇称的不可约表示，类C和类iC的特征标是一样的；对于奇宇称的不可约表示，类C和类iC的特征标差一个符号，偶宇称标记为g或+，奇宇称标记为u或−。

如果系统**初态为T_{2g}**对称性的态，则直积$\mathcal{H}'_{em} \otimes \psi_{T_{2g}}$包含可由取直积$\chi_{T_{1u}} \otimes \chi_{T_{2g}}$求得的不可约表示。

$\chi_{T_{1u}} \otimes \chi_{T_{2g}}$可由下表给出

Table 6.2. Characters for the direct product of the characters for the T_{1u} and T_{2g} irreducible representations of group O_h

E	$8C_3$	$3C_2$	$6C_2$	$6C_4$	i	$8iC_3$	$3iC_2$	$6iC_2$	$6iC_4$
9	0	1	−1	−1	−9	0	−1	1	1

左表A.31使用了上标+, − 来表达
偶（g）奇 (u) 宇称

利用分解公式得

$$a^{(T_{1u}\otimes T_{2g},A_{1g})}=\frac{1}{48}(1\times1\times9+8\times1\times0+3\times1\times1+6\times1\times(-1)+6\times1\times(-1)+1\times1\times(-9)$$
$$+8\times1\times0+3\times1\times(-1)+6\times1\times1+6\times1\times1)$$
$$=0$$

$$a^{(T_{1u}\otimes T_{2g},A_{2g})}=\frac{1}{48}(1\times1\times9+8\times1\times0+3\times1\times1+6\times(-1)\times(-1)+6\times(-1)\times(-1)+1\times1\times(-9)$$
$$+8\times1\times0+3\times1\times(-1)+6\times(-1)\times1+6\times(-1)\times1)$$
$$=0$$

$$a^{(T_{1u}\otimes T_{2g},E_g)}=\frac{1}{48}(1\times2\times9+8\times(-1)\times0+3\times2\times1+6\times0\times(-1)+6\times0\times(-1)+1\times2\times(-9)$$
$$+8\times(-1)\times0+3\times2\times(-1)+6\times0\times1+6\times0\times1)$$
$$=0$$

$$a^{(T_{1u}\otimes T_{2g},T_{1u})}=\frac{1}{48}(1\times3\times9+8\times0\times0+3\times(-1)\times1+6\times(-1)\times(-1)+6\times1\times(-1)+1\times(-3)\times(-9)$$
$$+8\times0\times0+3\times1\times(-1)+6\times1\times1+6\times(-1)\times1)$$
$$=1$$

$$a^{(T_{1u}\otimes T_{2g},T_{2u})}=\frac{1}{48}(1\times3\times9+8\times0\times0+3\times(-1)\times1+6\times1\times(-1)+6\times(-1)\times(-1)+1\times(-3)\times(-9)$$
$$+8\times0\times0+3\times1\times(-1)+6\times(-1)\times1+6\times1\times1)$$
$$=1$$

$$a^{(T_{1u}\otimes T_{2g},A_{1u})}=\frac{1}{48}(1\times1\times9+8\times1\times0+3\times1\times1+6\times1\times(-1)+6\times1\times(-1)+1\times(-1)\times(-9)$$
$$+8\times(-1)\times0+3\times(-1)\times(-1)+6\times(-1)\times1+6\times(-1)\times1)$$
$$=0$$

$$a^{(T_{1u}\otimes T_{2g},A_{2u})}=\frac{1}{48}(1\times1\times9+8\times1\times0+3\times1\times1+6\times(-1)\times(-1)+6\times(-1)\times(-1)+1\times(-1)\times(-9)$$
$$+8\times(-1)\times0+3\times(-1)\times(-1)+6\times1\times1+6\times1\times1)$$
$$=1$$
$$a^{(T_{1u}\otimes T_{2g},E_u)}=\frac{1}{48}(1\times2\times9+8\times(-1)\times0+3\times2\times1+6\times0\times(-1)+6\times0\times(-1)+1\times(-2)\times(-9)$$
$$+8\times1\times0+3\times(-2)\times(-1)+6\times0\times1+6\times0\times1)$$
$$=1$$
$$a^{(T_{1u}\otimes T_{2g},T_{1g})}=\frac{1}{48}(1\times3\times9+8\times0\times0+3\times(-1)\times1+6\times(-1)\times(-1)+6\times1\times(-1)+1\times3\times(-9)$$
$$+8\times0\times0+3\times(-1)\times(-1)+6\times(-1)\times1+6\times1\times1)$$
$$=0$$
$$a^{(T_{1u}\otimes T_{2g},T_{2g})}=\frac{1}{48}(1\times3\times9+8\times0\times0+3\times(-1)\times1+6\times1\times(-1)+6\times(-1)\times(-1)+1\times3\times(-9)$$
$$+8\times0\times0+3\times(-1)\times(-1)+6\times1\times1+6\times(-1)\times1)$$
$$=0$$

$$T_{1u}\otimes T_{2g}=A_{2u}+E_u+T_{1u}+T_{2u}$$

因此我们得电偶极跃迁从T_{2g}态的选择定则只能给出具有A_{2u},E_u,T_{1u},T_{2u}对称性的态。
而且，$\mathcal{H}'_{em}$是一个奇函数，电偶极跃迁仅能耦合宇称相反的态。

同样的论证可用于求立方对称性下任意初态和末态的选择定则:
例如:

$$E_g \otimes T_{1u} = T_{1u} + T_{2u}$$

$$T_{1u} \otimes T_{1u} = A_{1g} + E_g + T_{1g} + T_{2g}$$

T_{1u}：表示基函数(x, y, z)

Table 6.3. Character table for the pint group D_4 (422)

D_4 (422)			E	$C_2 = C_4^2$	$2C_4$	$2C_2'$	$2C_2''$
$x^2 + y^2, z^2$		A_1	1	1	1	1	1
	R_z, z	A_2	1	1	1	−1	−1
$x^2 - y^2$		B_1	1	1	−1	1	−1
xy		B_2	1	1	−1	−1	1
(xz, yz)	(x, y) (R_x, R_y)	E	2	−2	0	0	0

假设我们考虑将对称性从O_h降低到D_4，根据 D_4群和$S_2 = C_i$群的特征标表，我们可以给出直积群D_{4h}的特征标。我们注意到 D_{4h}对称性下的矢量并不随单个的不可约表示变换。

$$\left.\begin{array}{c} z \to A_{2u} \\ (x, y) \to E_u \end{array}\right\}$$

所以 O_h对称性的T_{1u}变成了D_{4h}对称性的$A_{2u} + E_u$。

Table 6.3. Character table for the pint group D_4 (422)

D_4 (422)			E	$C_2 = C_4^2$	$2C_4$	$2C_2'$	$2C_2''$
x^2+y^2, z^2		A_1	1	1	1	1	1
	R_z, z	A_2	1	1	1	−1	−1
x^2-y^2		B_1	1	1	−1	1	−1
xy		B_2	1	1	−1	−1	1
(xz, yz)	(x, y) (R_x, R_y)	E	2	−2	0	0	0

进而 O_h群中的T_{2g}变成了D_{4h}群中的$E_g + B_{2g}$。
因此对D_{4h}群，电偶极跃迁仅将A_{1g}态与E_u和A_{2u}态耦合起来，对于一个E_g态，根据它与矢量的直积有

$$E_g \otimes (A_{2u} + E_u) = E_g \otimes A_{2u} + E_g \otimes E_u$$
$$= E_u + (A_{1u} + A_{2u} + B_{1u} + B_{2u})$$

因此对于D_{4h}群，电偶极跃迁可以将E_g态和任意奇宇称态耦合起来。这说明当我们降低对称性的时候，选择定则的限制减弱了，从而允许更多的跃迁。

Q1: 如果已经定义了一个空间中的矢量$\vec{R}$，如何判断另外一个对象是否是矢量？

在这个空间里所包含的各种线性操作（operation）下，
它的变换跟$\vec{R}$一样就是矢量，否则就不是。

Q2: 什么叫拥有 XXX 对称性？

跟对称群 XXX 不可约表示的基函数一样变换！

Wigner-Eckart theorem

The early days of atomic spectroscopy saw a massive mess of confusion, to say the least. Many transition lines with varying intensity were observed, while some expected lines were missing. In quantum mechanics, the probability amplitude for a transition from some initial state $|i\rangle$ to some final state $|f\rangle$ due to some perturbation, such as the electromagnetic field, usually ends up being given by an operator $\mathcal{O}$ evaluated between the two states, that is, $\langle f|\,\mathcal{O}\,|i\rangle$.

In atomic spectroscopy, the initial and final states transform like members of some irreducible representations of the rotation group $SO(3)$, respectively, $|i\rangle = |\alpha, j, m\rangle$ and $|f\rangle = |\alpha', j', m'\rangle$. Here α and α' denote generically some other quantum numbers not governed by $SO(3)$, such as the principal quantum number that measures how "high" a given state is in the energy spectrum.

The operator $\mathcal{O}$ also transforms like members of some irreducible representations of $SO(3)$. We write $\mathcal{O}_{JM}$ to indicate that it transforms like the state $|JM\rangle$. For example, in the simplest kind of electromagnetic transition, known as the dipole transition, $\mathcal{O}$ is simply the position operator $\vec{x}$ of the electron. In this case, $\mathcal{O}$ transforms like a vector, and thus, as explained in chapter IV.2, $J = 1$ and $M = -1$, 0, or 1; in other words, $\mathcal{O}_{1M}$ transforms like the spherical harmonic Y_1^M.

The Wigner-Eckart theorem tells us what group theory has to say about the matrix element $\langle \alpha', j', m' \,|\, \mathcal{O}_{JM} \,|\, \alpha, j, m\rangle$. It states[3] that

$$\begin{aligned}\langle \alpha', j', m' \,|\, \mathcal{O}_{JM} \,|\, \alpha, j, m\rangle &= \Big(\big\langle j', m'\big|\ \big(\,|JM\rangle \otimes |j, m\rangle\,\big)\Big)\langle \alpha', j' \| O_J \| \alpha, j\rangle \\ &= \langle j', m' \,|\, J, j, M, m\rangle \langle \alpha', j' \| O_J \| \alpha, j\rangle\end{aligned} \tag{16}$$

The amplitude factors into a product of two quantities, which we can associate with symmetry and with dynamics. The "thing with the double vertical bars"[4] $\langle \alpha', j' \| O_J \| \alpha, j\rangle$, called the reduced matrix element of the operator $\mathcal{O}$, represents dynamics, about which group theory has nothing to say. Its evaluation requires knowing the Schrödinger wave functions of the initial and final states and doing an integral. One result of the theorem is that the quantity $\langle \alpha', j' \| O_J \| \alpha, j\rangle$ depends on α', α, J, j', and j but does not depend on m' and m.

Selection rules:

$$|\Delta j| = |j' - j| \leq J$$

$$\Delta m = m' - m = M \leq J$$

Clebsch-Gordan coefficeint

Clebsch–Gordan coefficients

$$(2j_1+1)(2j_2+1)$$

$$\uparrow$$

$$|j_1 m_1\rangle \otimes |j_2 m_2\rangle$$

$$J^{tot} = |j_1 - j_2|, |j_1 - j_2| + 1, \cdots, |j_1 + j_2|$$

$$\sum_{l=j_1-j_2}^{j_1+j_2} 2l + 1 = (2j_1+1)(2j_2+1)$$

The tensor product of these spaces, $V_3 \equiv V_1 \otimes V_2$, has a $(2\,j_1 + 1)\,(2\,j_2 + 1)$-dimensional *uncoupled* basis

$$\int_0^{\pi} \int_0^{2\pi} Y_{lm} Y^*_{l'm'}\, d\Omega = \delta_{ll'}\delta_{mm'} \quad (d\Omega = \sin\theta d\theta d\varphi)$$

A quick question:

since $\int_0^{\pi}\int_0^{2\pi} Y_{lm}Y^*_{l'm'}\,d\Omega = \delta_{ll'}\delta_{mm'}$,
then $\int_0^{\pi}\int_0^{2\pi} Y_{lm}\,d\Omega = ?$ In the case of $l \neq 0$

Answer:

$\int_0^{\pi}\int_0^{2\pi} Y_{lm}\,d\Omega \propto \int_0^{\pi}\int_0^{2\pi} Y_{lm}Y^*_{00}\,d\Omega = \delta_{l0}\delta_{m0} = 0$ due to $l \neq 0$

$Y_{00} = 1/\sqrt{4\pi}$



氢原子基态没有固有磁矩: $\int_0^{\pi}\int_0^{2\pi} Y_{lm}\,d\Omega \propto \int_0^{\pi}\int_0^{2\pi} Y_{00}Y_{lm}Y^*_{00}\,d\Omega \sim \langle 00|e\vec{r}|00\rangle = 0$

The Van der Waals Interaction

An important, nice application of the Rayleigh-Schrödinger perturbation theory is to calculate the long-range interaction, or **van der Waals force**, between two hydrogen atoms in their ground states. It is easy to show that the energy between the two atoms for large separation r is attractive and varies as r^{-6}.

Consider the two protons of the hydrogen atoms to be *fixed* at a distance r (along the z-axis) with $\mathbf{r}_1$ the vector from the first proton to its electron and $\mathbf{r}_2$ the vector from the second proton to its electron; see Figure 5.3. Then the Hamiltonian H can be written as

$$\begin{aligned} H &= H_0 + V \\ H_0 &= -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - \frac{e^2}{r_1} - \frac{e^2}{r_2} \\ V &= \frac{e^2}{r} + \frac{e^2}{|\mathbf{r}+\mathbf{r}_2-\mathbf{r}_1|} - \frac{e^2}{|\mathbf{r}+\mathbf{r}_2|} - \frac{e^2}{|\mathbf{r}-\mathbf{r}_1|}. \end{aligned} \tag{5.3.53}$$

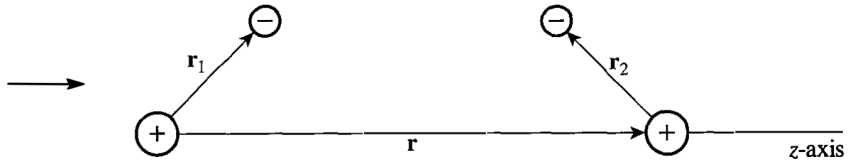


FIGURE 5.3 Two hydrogen atoms with their protons (+) separated by a fixed distance r and their electrons (−) at displacements $\mathbf{r}_i$ from them.

The lowest-energy solution of H_0 is simply the product of the ground-state wave functions of the noninteracting hydrogen atoms

$$U_0^{(0)} = U_{100}^{(0)}(\mathbf{r}_1)U_{100}^{(0)}(\mathbf{r}_2). \tag{5.3.54}$$

Now for large r ($\gg$ the Bohr radius a_0), expand the perturbation V in powers of $\mathbf{r}_i/\mathbf{r}$ to obtain

$$V = \frac{e^2}{r^3}(x_1x_2 + y_1y_2 - 2z_1z_2) + 0\left(\frac{1}{r^4}\right) + \cdots. \tag{5.3.55}$$

The lowest-order r^{-3}-term in (5.3.55) corresponds to the interaction of two electric dipoles $e\mathbf{r}_1$ and $e\mathbf{r}_2$ separated by $\mathbf{r}$. The higher-order terms represent higher-order multipole interactions, and thus every term in V involves spherical harmonics Y_l^m with $l_i > 0$ for each hydrogen atom. Hence, for each term in (5.3.55) the first-order perturbation energy matrix element $V_{00} \simeq 0$, because the ground state $U_0^{(0)}$ wave function (5.3.54) has $l_i = 0$ and $\int d\Omega Y_l^m(\Omega) = 0$ for l and $m \neq 0$). The second-order perturbation

$$E^{(2)}(r) = \frac{e^4}{r^6}\sum_{k\neq 0}\frac{|\langle k^{(0)}|x_1x_2 + y_1y_2 - 2z_1z_2|0^{(0)}\rangle|^2}{E_0^{(0)} - E_k^{(0)}} \tag{5.3.56}$$

will be nonvanishing. We immediately see that this interaction varies as $1/r^6$; since $E_k^{(0)} > E_0^{(0)}$, it is negative. This $1/r^6$ long-range attractive van der Waals potential is a general property of the interaction between two atoms in their ground state.*

Vector operator $\vec{V}$

Defining property : $[V_i, L_j] = i\epsilon_{ijk}\hbar V_k$

when $\vec{V} \to \vec{L}$, Lie algebra

You can check $[x, L_z] = -i\hbar y$, $[p_x, L_z] = -i\hbar p_y$, etc.

Spherical tensors:

$$T_m^{(l)} = Y_m^l(\hat{n})$$

$$Y_0^1 = \sqrt{3/4\pi}\cos\theta = \frac{z}{r} \rightarrow T_0^1 = \sqrt{3/4\pi}V_z$$

$$Y_{\pm 1}^1 \propto \mp\sqrt{3/4\pi}\,\frac{x \pm iy}{\sqrt{2}r} \rightarrow T_{\pm}^1 = \sqrt{3/4\pi}\left(\mp\frac{V_x \pm iV_y}{\sqrt{2}r}\right)$$

- $\vec{P}$ 与 $\vec{R}$ 以及电偶极矩一样，是矢量表示。

对照球谐张量， $T_m^{(l)}: 2l+1$ components.

For $\vec{P}, l=1$

$$P_x = 1/\sqrt{2}(P_1^{-1} - P_{-1}^1)$$
$$P_y = i/\sqrt{2}(P_1^1 + P_{-1}^1)$$
$$P_z = P_0^1$$

- $\vec{R} = \left((-x - iy)/\sqrt{2}, z, (x - iy)/\sqrt{2}\right)$ is a rank-1 spherical tensor, from the Wigner-Eckart theorem, the same discussion holds as the momentum vector $\vec{P}$.

Another perspective: **Parity**

($\Delta l \neq 0$)

Spherically symmetric Schrodinger equation under parity conservation

The angular solutions to the Schrödinger equation for a spherically symmetric potential with no spin are the usual spherical harmonics.

$$Y_{lm}(\theta,\phi)=\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l^m(\cos\theta)\exp(im\phi)$$

under the parity operation :

$$\theta\to\pi-\theta \quad (\cos\theta\to-\cos\theta)$$
$$\phi\to\pi+\phi$$

Since lth **Legendre polynomial**, defined by the **Rodrigues formula**:

$$P_l(x)=\frac{1}{2^l l!}\frac{d^l}{dx^l}[(x^2-1)^l]$$

associated Legendre function： （$m\geq 0$）

$$P_l^m(x)=(-1)^m(1-x^2)^{m/2}\frac{d^m P_l(x)}{dx^m}$$

$$P_l^{-m}(x)=(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x)$$

$$Y_{l,-m}(\theta,\phi)=(-1)^m Y_{l,m}(\theta,\phi)^*$$

Spherically symmetric Schrodinger equation under parity conservation

$(x^2-1)^l$ is an even function of x,

$$P_l^m(-\cos\theta) = (-1)^{l+m} P_l^m(\cos\theta)$$

$$\exp\big(im(\pi+\phi)\big) = (-1)^m \exp(im\phi)$$

and we have

$$Y_{lm} \to (-1)^{l+m}(-1)^m Y_{lm} = (-1)^l Y_{lm}$$

Thus :

$s, d, g\cdots$ waves have even parity.

$p, f, h\cdots$ waves have odd parity .

$$\langle n', l', m' | \vec{A}.\vec{P} | n, l, m \rangle$$

When $l = l'$, this term vanishes.

Parity is not conserved.

Selection rule: $\Delta l = l_f - l_i, \Delta m = m_f - m_i$ (final/initial state : f, i)

$$P_x : \Delta l = \pm 1, \Delta m = \pm 1$$
$$P_y : \Delta l = \pm 1, \Delta m = \pm 1$$
$$P_z : \Delta l = \pm 1, \Delta m = 0$$

Why do we need to discuss parity, it is not in $SO(3)$?

- It is not !!!

Parity $i = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \det(i) = -1$

$$i \notin SO(3)$$

$$SO(3) + iSO(3) = O(3), \qquad \frac{O(3)}{SO(3)} = \mathbb{Z}_2 = \{e, i\}$$

Selection rules

- $|n,l,m\rangle$ the basis for hydrogen atom

Examples: (make a quick guess)

$\langle 3,1,1|\ P_z|2,1,0\rangle$ $\langle 3,1,1|\ P_z|2,1,1\rangle$

$\langle 3,2,1|\ P_x|2,0,0\rangle$ $\langle 3,1,1|\ P_x|2,0,0\rangle$

No constraints on the principle quantum number n